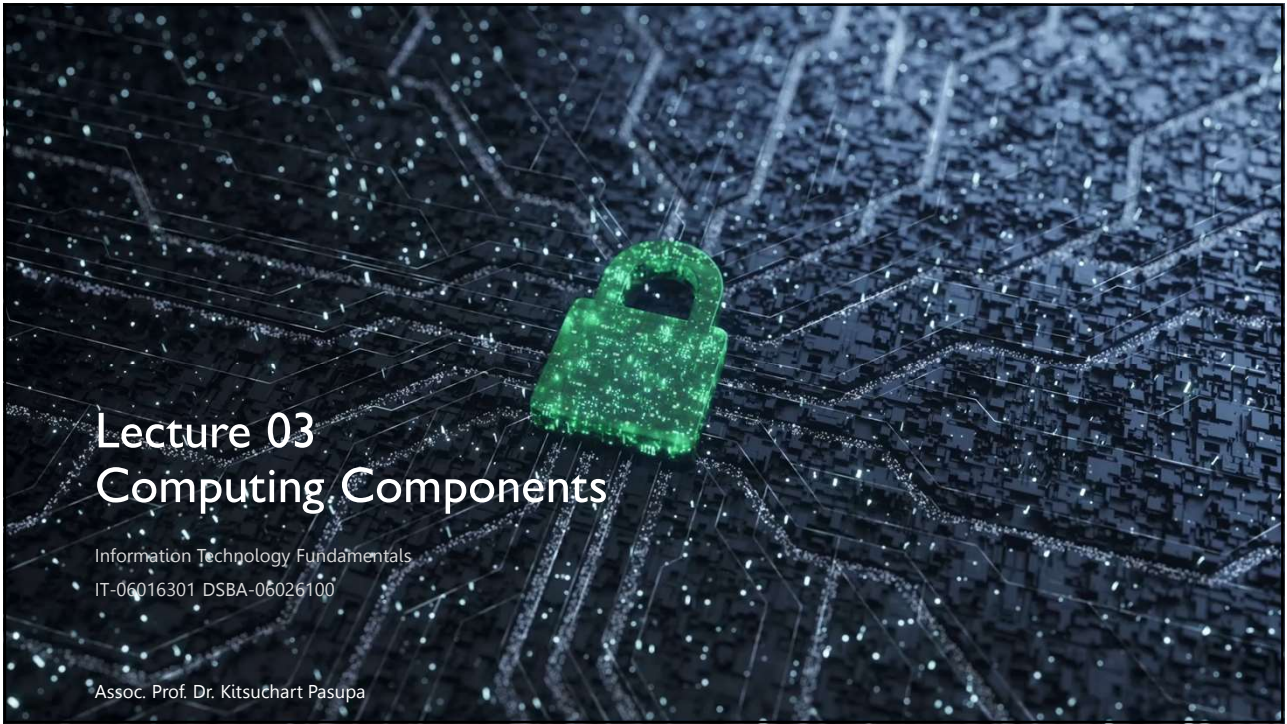


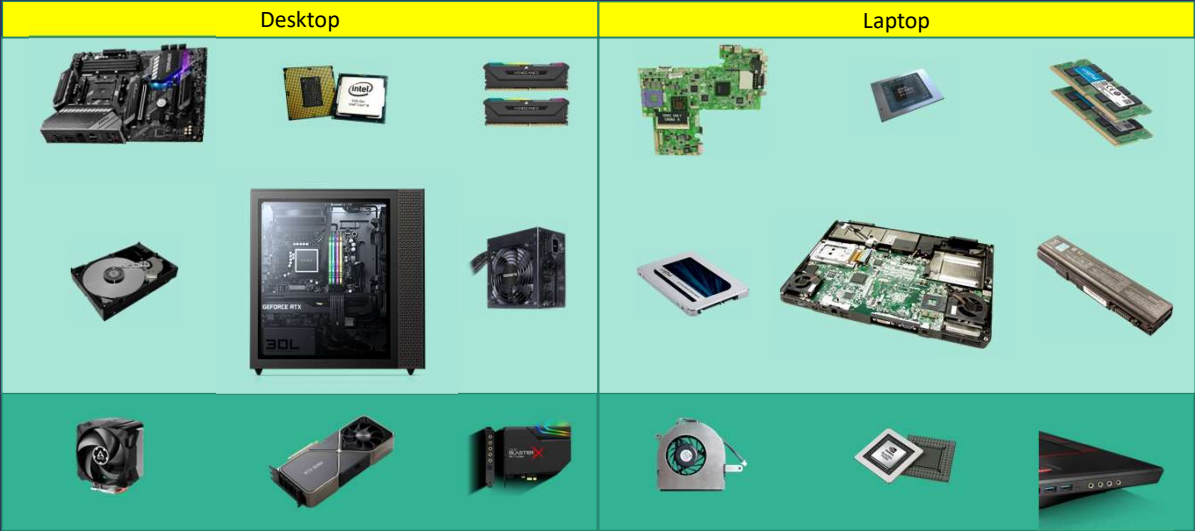


1



2

Inside the case



3

Inside the case



4

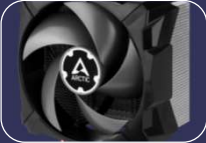
Inside the case



Motherboard



Processor



Cooling Device



Memory



Adapter



Power Supply



Storage

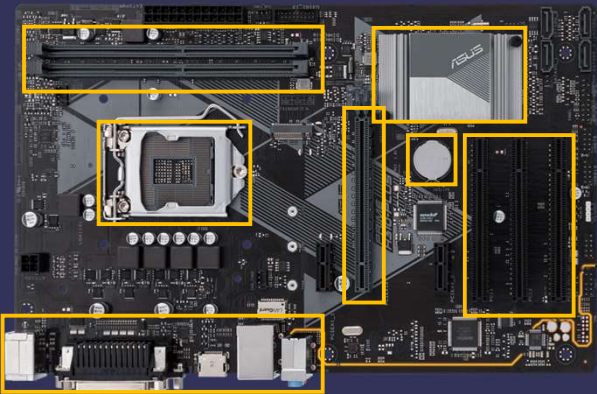
5

Motherboard

Slot for Memory Modules

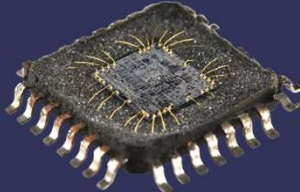
Slot for Processor Chip

The motherboard is the main circuit board of the computer



Chipset+Heatsink

CMOS Battery



Ports

Slots for Adapter Cards

A computer chip contains integrated circuits

6

Inside the case



Motherboard



Processor



Cooling Device



Memory



Adapter



Power Supply



Storage

7

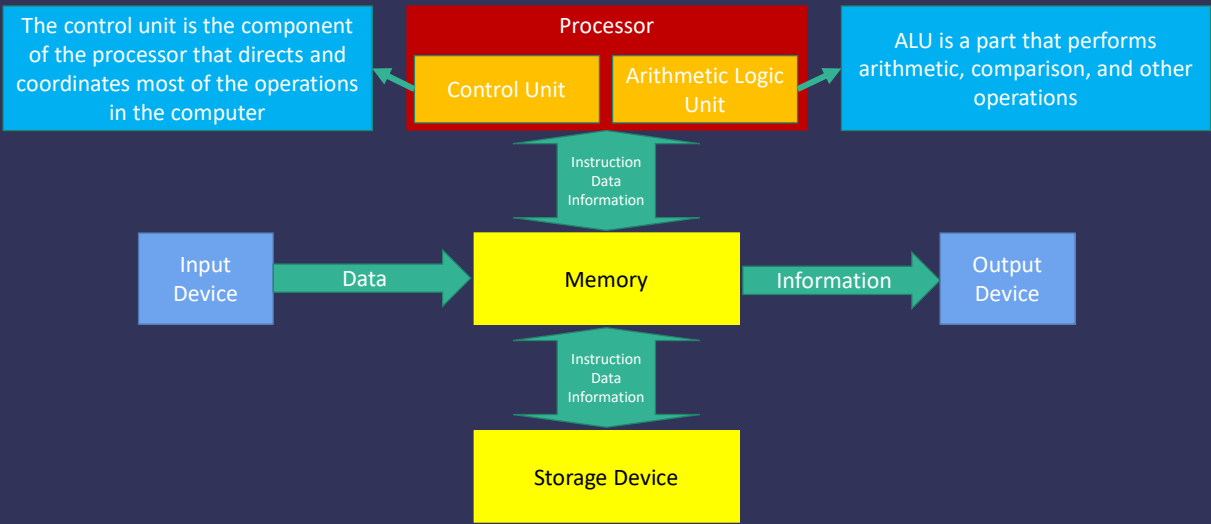
Central Processing Unit

- The central processing unit (CPU), also known as the processor, interprets and carries out the basic instructions that operate a computer
- A single CPU performs as a multi-core processor
 - Two or more separate cores in a single one
- It consists of a control unit and an arithmetic logic unit (ALU)



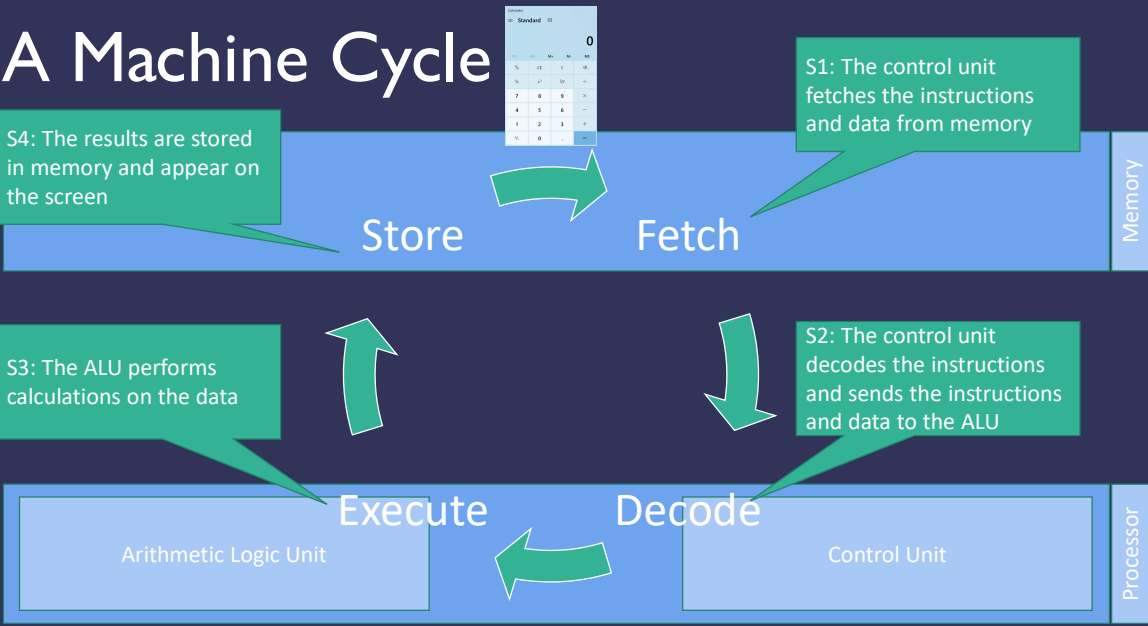
8

Central Processing Unit



9

A Machine Cycle



10

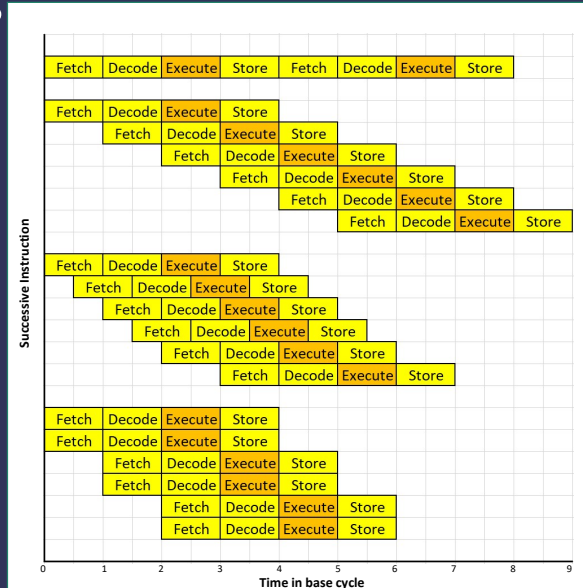
Instruction Pipelining

No Pipelining

Pipelining: Processor begins fetching a second instruction before it completes the machine cycle for the first instruction. Perform only one stage per clock cycle

Super-pipelined system is capable of performing two pipeline stages per clock cycle

Superscalar performs only one pipeline stage per clock cycle in each parallel pipeline

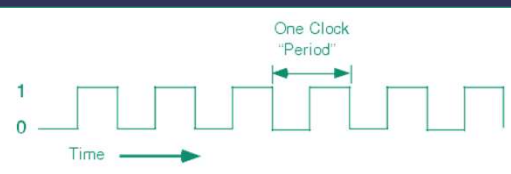


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Processor

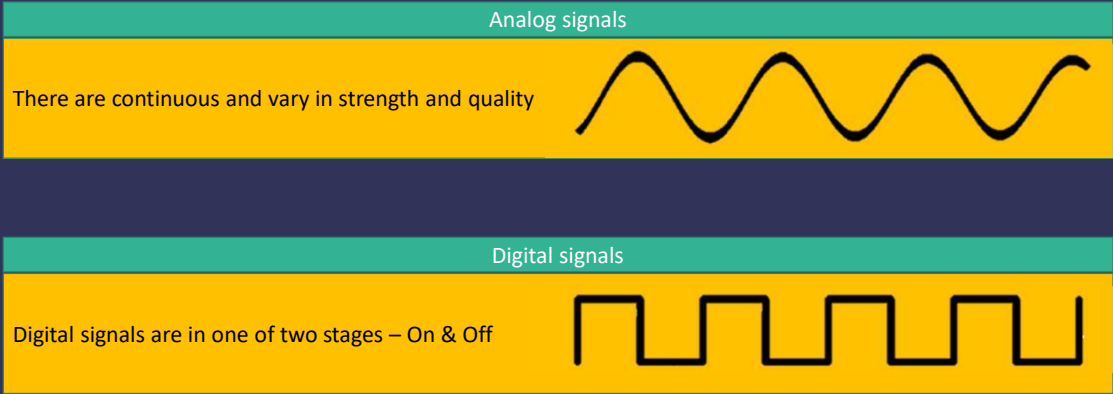
- The processor contains registers, that temporarily hold data and instructions
 - Register is one of a small set of data holding places
 - It is a quickly accessible location
 - Having adequate CPU register will help in faster and smooth execution
- The system clock controls the timing of all computer operations
 - The pace of the system clock is called the clock speed, and is usually measured in gigahertz

1 GHz = 1,000,000,000 cycle/second



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Data Representation



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Data Representation

Most computers are digital that is the binary system
It is all 0's and 1's in the computer. We call them "bits"

Using two unique digits (0 and 1)
8 bits = 1 byte = 1 character



Characters								
2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰	DEC
0	1	0	0	0	1	0	1	69
0	0	1	0	1	0	1	0	42
0	0	1	1	0	1	1	0	54

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ASCII Table

American Standard Code For Information Interchange

CHAR	DEC
E	69
*	42
6	54

Decimal	Hexadecimal	Binary	Octal	Char	Decimal	Hexadecimal	Binary	Octal	Char	Decimal	Hexadecimal	Binary	Octal	Char
0	0	0	0	(NUL)	48	30	110000	60	G	96	60	1100000	140	g
1	1	1	1	(START OF HEADING)	49	31	110001	61	g	97	61	1100001	141	a
2	2	10	2	(START OF TEXT)	50	32	110010	62	2	98	62	1100010	142	b
3	3	11	3	(END OF TEXT)	51	33	110011	63	3	99	63	1100011	143	c
4	4	100	4	(END OF TRANSMISSION)	52	34	110100	64	4	100	64	1100100	144	d
5	5	101	5	(ENQUIRY)	53	35	110101	65	5	101	65	1100101	145	e
6	6	110	6	(ACKNOWLEDGE)	54	36	110110	66	6	102	66	1100110	146	f
7	7	111	7	(BEL)	55	37	110111	67	7	103	67	1100111	147	h
8	8	1000	10	(BACKSPACE)	56	38	111000	70	8	104	68	1101000	150	h
9	9	1001	11	(HORIZONTAL TAB)	57	39	111001	71	9	105	69	1101001	151	i
10	A	1010	12	(LINE FEED)	58	3A	111010	72	1	106	6A	1101010	152	j
11	B	1011	13	(VERTICAL TAB)	59	3B	111011	73	1	107	6B	1101011	153	k
12	C	1100	14	(FORM FEED)	60	3C	111100	74	0	108	6C	1101100	154	l
13	D	1101	15	(CARRIAGE RETURN)	61	3D	111101	75	0	109	6D	1101101	155	m
14	E	1110	16	(SHIFT OUT)	62	3E	111110	76	0	110	6E	1101110	156	n
15	F	1111	17	(SHIFT IN)	63	3F	111111	77	1	111	6F	1101111	157	o
16	10	10000	20	(DATA LINK ESCAPE)	64	40	1000000	100	@	112	70	1110000	160	p
17	11	10001	21	(DEVICE CONTROL 1)	65	41	1000001	101	A	113	71	1110001	161	q
18	12	10010	22	(DEVICE CONTROL 2)	66	42	1000010	102	B	114	72	1110010	162	r
19	13	10011	23	(DEVICE CONTROL 3)	67	43	1000011	103	C	115	73	1110011	163	s
20	14	10100	24	(DEVICE CONTROL 4)	68	44	1000100	104	D	116	74	1110100	164	t
21	15	10101	25	(NEGATIVE ACKNOWLEDGE)	69	45	1000101	105	E	117	75	1110101	165	u
22	16	10110	26	(SYNCHRONOUS IDLE)	70	46	1000110	106	F	118	76	1110110	166	v
23	17	10111	27	(END OF TRANS. BLOCK)	71	47	1000111	107	G	119	77	1110111	167	w
24	18	11000	30	(CANCEL)	72	48	1001000	110	H	120	78	1111000	170	x
25	19	11001	31	(END OF MEDIUM)	73	49	1001001	111	I	121	79	1111001	171	y
26	1A	11010	32	(SUBSTITUTE)	74	4A	1001010	112	J	122	7A	1111010	172	z
27	1B	11011	33	(ESCAPE)	75	4B	1001011	113	K	123	7B	1111011	173	{
28	1C	11100	34	(FILE SEPARATOR)	76	4C	1001100	114	L	124	7C	1111100	174	
29	1D	11101	35	(GROUP SEPARATOR)	77	4D	1001101	115	M	125	7D	1111101	175	}
30	1E	11110	36	(RECORD SEPARATOR)	78	4E	1001110	116	N	126	7E	1111110	176	~
31	1F	11111	37	(UNIT SEPARATOR)	79	4F	1001111	117	O	127	7F	1111111	177	[DEL]
32	20	100000	40	(SPACE)	80	50	1010000	120	P					
33	21	100001	41	!	81	51	1010001	121	Q					
34	22	100010	42	"	82	52	1010010	122	R					
35	23	100011	43	#	83	53	1010011	123	S					
36	24	100100	44	\$	84	54	1010100	124	T					
37	25	100101	45	%	85	55	1010101	125	U					
38	26	100110	46	&	86	56	1010110	126	V					
39	27	100111	47	'	87	57	1010111	127	W					
40	28	101000	50	(	88	58	1011000	130	X					
41	29	101001	51	)	89	59	1011001	131	Y					
42	2A	101010	52	*	90	5A	1011010	132	Z					
43	2B	101011	53	+	91	5B	1011011	133	[					
44	2C	101100	54	,	92	5C	1011100	134	\					
45	2D	101101	55	-	93	5D	1011101	135	]					
46	2E	101110	56	.	94	5E	1011110	136	^					
47	2F	101111	57	/	95	5F	1011111	137	_					

<https://commons.wikimedia.org/wiki/File:ASCII-Table.svg>

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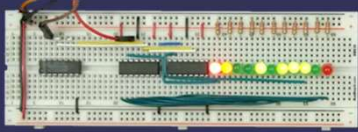
How is a letter converted to a binary form?

A user presses a key on a keyboard

The scan code for the letter is sent to the electronic circuitry in the computer



The electronic circuitry in the computer converts the scan code for the letter to its ASCII binary code (00110101) and store it in memory for processing



Binary							Hex
0	0	1	0	1	1	1	2E
Binary							Dec
0	0	1	1	0	1	0	53

<https://www.youtube.com/watch?v=7aXbh9VUB3U>

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Inside the case



Motherboard



Processor



Cooling Device



Memory



Adapter



Power Supply



Storage

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Cooling Device

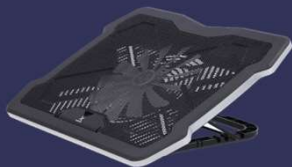
A processor chip generates heat that could cause the chip to malfunction or fail. Cooling device is necessary.



Heat Sink



Liquid Cooling Technology



Cooling Pad

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Inside the case



Motherboard



Processor



Cooling Device



Memory



Adapter



Power Supply



Storage

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Memory

- Memory consists of electronic components that store instructions waiting to be executed by the processor, data needed by those instructions, and the results of processing the data
- Each location in memory has an address

Volatile Memory

Loses its contents when power is turned off, for examples, RAM

Non-volatile Memory

Does not lose contents when power is removed, for examples, ROM, flash memory, CMOS

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Volatile Memory - Read Access Memory (RAM)

Static RAM (SRAM)

SRAM is a semiconductor memory.

It stores data for as long as power is supplied in the system.

It is much faster and consume less power than DRAM.

It is mostly used as a cache memory for the CPU.



Dynamic RAM (DRAM)

DRAM is made up of a transistor and a capacitor within an integrated circuit. Data bits are stored in the capacitor.

It is much cheaper than SRAM. DRAM has a higher storage capacity

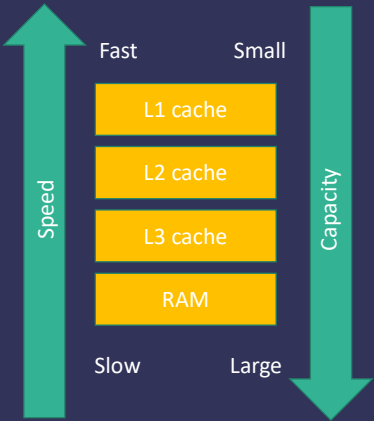
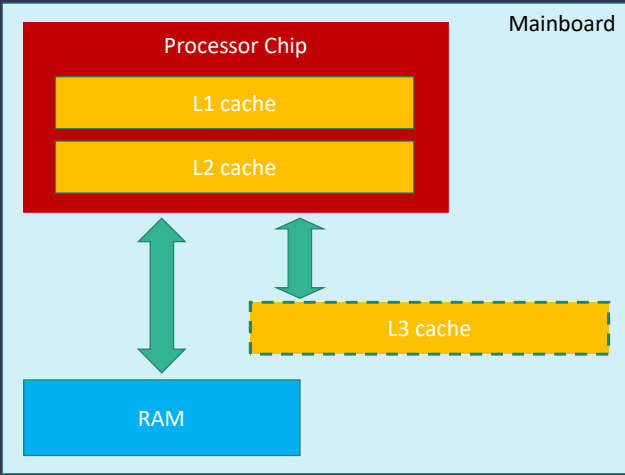
It is widely used as a computer's main memory.



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Memory Cache

- It speeds the processes of the computer because it stores frequently used instructions and data



22

Non-volatile Memory - Read Only Memory (ROM)

ROM

It refers to memory chips storing permanent data and instructions by manufacturer.

The microprocessor can read from ROM, but it cannot write to it or modify it.

It contain special instructions for the computer, that are always there because they're not erasable.

BIOS

It stands for Basic Input/Output System, pronounced as bye-oss.

It is a ROM chip found on motherboards that allows user to access and set up their computer system at the most basic level.

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Electrically Erasable Programmable Read-Only Memory (EEPROM)



It is called “flash memory”

It can be erased electronically and rewritten

Most computers use flash memory to hold their start-up instructions because it allows the computer to update its contents easily.

Flash memory chips store data and programs on many mobile devices and peripheral devices, such as smartphones, portable media players.

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Complementary Metal–Oxide–Semiconductor (CMOS) technology

It uses battery power to retain information when the power to the computer is off.

- Computer settings

Low consuming power

Slower than RAM

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Access Time

Access time is the amount of time it takes the computer processor to read data from the memory.

The access time of memory may be represented in either ns (nanoseconds).

The lower the time, the better or faster the computer should perform.

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Inside the case



Motherboard



Processor



Cooling Device



Memory



Adapter



Power Supply



Storage

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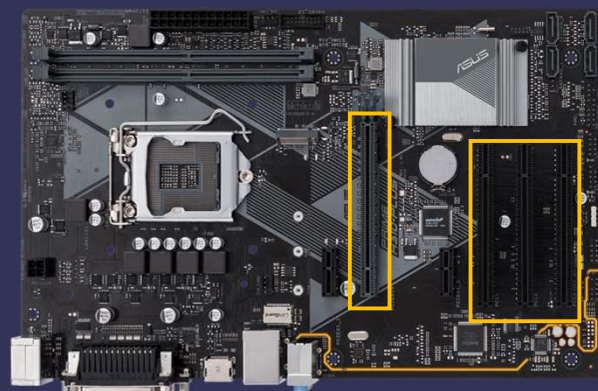
Adapters

Adapter Card

It enhances functions of a component of a desktop or system unit and/or provides connections to peripherals

Expansion slot

It is a socket on a motherboard that hold an adapter card. There are a few expansion slots (Peripheral Component Interconnect: PCI) in a motherboard.



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Adapter Cards

It comes with “Plug and Play” capability – the computer automatically can recognize peripheral devices as you install them



Graphics Processing Unit



Sound Card

29

Adapter

Dongle

A USB adapter enhances functions of a mobile computer and/or provides connections to peripheral devices



Bluetooth



Mobile Broadband Modem

30

Buses

Bus	Physical Types of Buses
It allows the various devices both inside and attached to the system unit to communicate with each other	System bus (Front-side bus): it connects processor to main memory
Data bus – transfers only data	Backside bus: it connect processor to cache
Address bus – transfers only address that store data	Expansion bus: it connect processor to peripherals
The width (word size) indicates the number of bit that can be transmitted at once. A 32-bit bus will share 32 bits of data. That is equivalent to 4 bytes (1 byte = 8-bit)	

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Inside the case



Motherboard



Processor



Cooling Device



Memory



Adapter



Power Supply



Storage

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Power Supply

- The power supply or laptop AC adapter converts the wall outlet AC power into DC power
- Mobile computers and devices can run using either a power supply or batteries
 - Batteries typically are rechargeable lithium-ion batteries



Desktop



Laptop

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Inside the case



Motherboard



Processor



Cooling Device



Memory



Adapter



Power Supply



Storage

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